



Supporting online learners' regulation skills with the help of learning analytics and generative artificial intelligence

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Abstract

Students often struggle to stay engaged and effectively regulate their learning in online environments, which can negatively impact their learning experiences. Despite the established importance of self-regulation of learning skills (SRLs) in maintaining engagement, many students face significant challenges in developing and implementing these skills due to a lack of adequate feedback. This is primarily due to tutors' high workloads and the difficulties inherent in engaging students in online settings. This study examines the impact of Learning Analytics (LA)-driven interventions to improve students' SRLs in online learning environments. Specifically, it compares the impact of feedback from Generative Artificial Intelligence (GenAI) and human tutors in a nine-week statistics course delivered via MOODLE at a higher education institution, employing LA and clustering techniques to model SRLs based on Ye & Pennisi's (2022) framework. In a quasi-experimental design, participants with varying SRLs were assigned to either a tutor-feedback or GenAI-feedback group. Feedback readability and reliability evaluations indicate that LA-driven GenAI-produced feedback was significantly more readable than human tutor feedback ($p < 0.01$) and demonstrated higher reliability than tutor-generated feedback. Results show that students in the low SRLs cluster receiving GenAI feedback exhibited statistically significant improvements in goal-setting skills ($p < 0.05$) and overall SRLs levels ($p < 0.05$) compared to the tutor-feedback group. In contrast, no significant differences were observed among high SRLs cluster students. This study underscores the potential of LA-driven GenAI feedback to be able to develop tailored, scalable feedback, improving SRLs performance of low SRLs students in online higher education contexts. Future research should explore these effects across diverse student groups and investigate the collaborative potential of semi-automated feedback systems that include tutors.

Keywords Self-regulated learning (SRL) · Generative artificial intelligence · Feedback · Learning analytics

Introduction

Online learning environments inherently provide learners with greater autonomy, requiring students to take more responsibility for managing their learning processes (Afacan Adanir & Gülbahar Güven, 2023; Lehmann et al., 2014). However, this autonomy is effective only when supported by strong self-regulated learning skills (SRLs) (Kizilcec et al., 2017). SRL encompasses cognitive, metacognitive, and motivational activities, such as attending to instruction, integrating knowledge, and fostering positive beliefs about learning capabilities (Schunk, 2001). This study focuses on the critical SRLs that learners use to manage and regulate their learning in online environments. Specifically, it examines goal setting (GS), task strategies (TS), time management (TM), help-seeking (HS), and self-evaluation (SE) skills that have been both theoretically discussed and evidenced empirically as important for improving self-regulation and eventually contributing to a more efficacious learning experience in online learning contexts (Barnard et al., 2009).

Research shows that students with low SRLs often struggle with course engagement (Tang, 2021) and academic performance (Lee et al., 2008; Martin et al., 2022). Without robust SRLs, students may face significant challenges navigating their learning process (Lajoie & Azevedo, 2006). Conversely, learners proficient in SRLs can successfully achieve their educational goals, even in less structured settings (Kizilcec et al., 2017). Given the challenges online learners face in developing and practising SRLs (Edisherashvili et al., 2022), there is a critical need for external support (Bjork et al., 2013; Zheng et al., 2015). While traditional classroom settings, where instructors can provide immediate guidance, online environments require tailored strategies for fostering SRLs (Barz et al., 2024). However, tutor-generated external supports, such as responses and feedback, face scalability issues due to the increased number of learners and the resource-intensive nature of such interventions (Dai et al., 2024; Fu et al., 2015; Nazaretsky et al., 2024; Wiley, 2006).

Learning Analytics (LA)-based solutions have become increasingly prominent as they involve the collection and analysis of learner data to optimise learning experiences (Gasovic et al., 2015). LA can help monitor SRLs by tracking student behaviour and providing a basis for timely interventions (Schmitz & Wiese, 2006; Zimmerman, 2008). LA-based written feedback interventions have shown promise in enhancing SRLs by providing data-driven insights (Chen, 2024; Lim et al., 2021; Tsai et al., 2021). However, several challenges persist. Existing LA-based written feedback interventions, such as OnTask (Tsai et al., 2021), are rule-based, meaning they are pre-designed and limited in their ability to adapt dynamically to learners' evolving needs (Garrison, 1995). Additionally, the scalability of LA-based written feedback remains challenging since current examples still rely heavily on human tutors for feedback provision (Pardo et al., 2019).

Recent advancements in Generative Artificial Intelligence (GenAI), particularly Large Language Models (LLMs) such as ChatGPT, offer the potential to augment feedback systems, making them more scalable and personalised (Bozkurt, 2023; Zhou et al., 2024). Despite this potential, there is a notable lack of empirical research exploring GenAI's impact on student learning (Lan & Zhou, 2025), particularly within LA interventions. Most studies to date are conceptual (Chiu et al., 2023) rather than experimental, leaving critical questions about GenAI's effectiveness in supporting feedback interventions for the development of SRLs in online learning settings unanswered. Existing research has primarily focused on AI's ability to provide feedback on student writing (Escalante et al., 2023; Meyer et al., 2024; Weber et al., 2024). However,

a significant gap remains in understanding how AI-generated feedback compares to feedback provided by human tutors, especially in relation to SRLs in online learning contexts. Thus, we aim to evaluate LA-empowered GenAI-based written feedback as a practical solution for enhancing feedback provision on SRLs in online learning settings, diagnosing SRLs based on observed behaviours in Learning Management System (LMS) interactions. Specifically, we examine both tutor-generated and AI-generated feedback in fostering SRLs, aiming to explore their respective strengths and gain insights into how each can support students' SRLs in online learning.

This study is specifically guided by the following three research questions:

RQ1: What SRLs profiles emerge from analysing students' LMS log data?

RQ2: To what extent is GenAI-generated feedback readable and reliable compared to tutor-generated feedback on observed SRLs behaviours?

RQ3: How do tutor and GenAI-generated feedback interventions impact different levels of students' SRLs behaviours?

By addressing the above research questions, we aim to clarify the strengths and limitations of using LA-empowered GenAI to provide feedback on students' SRLs.

Given the growing diversity of today's online education settings, research into technology-based solutions can help address students' tailored needs, enabling educators with limited resources to provide personalised support. Moreover, such findings are critical for guiding the development of fair and equitable online learning environments. By adopting potential strategies, educational institutions can foster more inclusive and applicable learning practices for all learners.

Literature review

The emergence of LA in SRLs

With the idea of formative assessment, it has become possible to support learners during the development stage and promote their performance. This perspective has reshaped approaches to collecting and utilising educational data (Ye, 2022). Traditional methods, such as questionnaires, rely on students' self-reports and are typically administered at appointed points (e.g., the beginning or end of a course). Although they offer valuable insights, these methods are limited in their ability to capture real-time processes that involve skills like SRLs. They often lack the sensitivity required to detect changes in students' self-regulation over time (Du et al., 2023). As a result, there is an increasing demand for approaches that enable continuous, real-time monitoring of learners' behaviours. LA addresses this need by examining the behavioural traces learners leave in computer-based environments, thereby offering a more dynamic insight into learning processes. By leveraging LA, researchers can obtain representations of SRL in action (Bernacki, 2017; Saqr et al., 2023) and effectively track how students' regulation skills evolve during task engagement (Van Halem et al., 2020). Consequently, LA has become a key instrument for supporting skills such as SRL that demand real-time observation and timely intervention (Du et al., 2023). Accordingly, this study adopted an LA-based approach to investigate and gain timely insights into online students' SRLs.

LA-based feedback to support SRLs

In recent studies on LA-based investigations of SRLs, researchers often use interventions through various forms of feedback, from data visualisations to written feedback. Numerous studies indicate the limitations of data visualisations and show that students consider written feedback to be a particularly valuable method of receiving feedback (Ferguson, 2011; Taylor & Burke Da Silva, 2014; Voelkel et al., 2020). However, written feedback interventions on LA diagnosis of SRLs are scarce in the literature. For instance, Lim et al. (2021) explored the influence of LA-driven written feedback on students' SRLs and academic performance within a cohort in a flipped classroom setting. The objective was to evaluate the efficacy of feedback interventions on SRLs in online learning tasks. The study utilised log data and e-books to provide timely, rule-based written feedback to students in the experimental group over a semester. Based on students' task assessments, two distinct written feedback interventions were delivered in the fifth and ninth weeks to improve their SRLs. The quantitative findings demonstrated that the feedback positively impacted students' SRLs.

Although this study provides insight into the impact of LA-based written feedback on SRLs, one key limitation is the reliance on raw data processing, which often fails to capture the complexities of students' learning trajectories. The study conducted by Lim et al. (2021) highlights the need for fine-grained, processed data to provide a deeper understanding of learner behaviours and progress (Alhazbi et al., 2024). This limitation aligns with the view that LA studies often lack robust theoretical frameworks for data collection and analysis, underscoring the need for enhanced methodological rigour in future research (Fan et al., 2021; Mangaroska & Giannakos, 2019; Ye & Pennisi, 2022). To address this limitation, the present study adopts a theory-based framework (please see Table 1 for details) proposed by Ye and Pennisi (2022), specifically tailored to SRL. This framework integrates Pintrich's (2000) model of SRL, containing forethought, planning, activation, monitoring, control, and reflection, with the Online Self-Regulated Learning Questionnaire (OSLQ; Barnard et al., 2009), as a foundational lens for data collection. Ye and Pennisi's (2022) framework was chosen because the SRL skills indicators it provides are aligned with behavioural traces that can be captured from online learning platforms, enabling the assessment of learners' SRL skills across all phases of the SRL cycle. Furthermore, the course resources from which our data were collected closely correspond to the indicators specified in this framework, ensuring both conceptual and practical alignment between our data sources and the measurement model. Thus, the framework bridges theory, measurement, and platform-specific behavioural indicators, making it the most applicable and practical choice for the objectives of this study.

Exploring GenAI in SRL

With the rapid advancement of GenAI, research on its educational applications has grown steadily, ranging from writing support (Meyer et al., 2024) to mathematics learning (Kumar et al., 2025). Beyond task-specific gains, recent reviews highlight its potential for fostering SRL. Sardi et al. (2025) reported that most studies indicate positive effects of GenAI on SRL, primarily through adaptive and personalised feedback. However, they also mentioned several unexplored issues, including a lack of evidence on the long-term impact and the still underexplored role of tutors in AI-supported learning scenarios. In their synthesis of

Table 1 Matching map between the SRL phases, learning behavioural data in LMS, and OSLQ (Ye & Pen-nisi, 2022, p.3)

PINTRICH'S SRL MODEL PHASE	LEARNING BEHAVIOURAL VARIABLES IN LMS (CODE)	OSLQ SCALE (CODE)
FORETHOUGHT, PLANNING, AND ACTIVATION	Time spent on reviewing the syllabus (SyTim) and the rubrics (RubTim); if the syllabus has been downloaded (SyIDow) Syllabus (SyIFre) and rubrics (RubFre) visit frequency	Goal setting (GS)
MONITORING	Course logins (LoginM), total item visits (TotVisM), topic visits (TopVisM) per module Average days accessing the course per module (DayVisM)	Task strategies (TS)
CONTROL	Average time spent on each module (TimeM) and additional resources (AddRes) Number of threads (#DisThr) and replies (#DisRep) created in online discussions; number of posts read (#PostRea) Lecture completion rate (LecCom) Average number of late submissions per module (LatSubAv); Average hours submitting assignments before the deadline (Advance) Average module completion rate before the deadline (MComAv)	Time management (TM)
REACTION AND REFLECTION	Number of total questions asking to the instructor(s) (#QSum) and asking for an extension (#AskExt) Number of times asking the instructor (s) questions excluding asking for extension (#QAskins)	Help-seeking (HS)
	Average number of revisiting a module after the grades are published (Revisit) Optional self-assessments (SAssCom) and bonus quiz (BQCom) completion rate Average number of retaking quizzes (RetakeQ)	Self-evaluation (SE)

*While the OSLQ framework incorporates an Environmental Structure pillar, it lacks a corresponding log indicator and therefore does not include it in its analysis.

14 studies, Lan and Zhou (2025) highlighted that most interventions focused on isolated phases of the SRL cycle, leaving other phases insufficiently examined. They also noted instances where AI-generated feedback did not align with learners' needs, emphasising the importance of grounding feedback in reliable sources, such as real-time LA data. Taken together, these reviews caution against overreliance on GenAI support. Although learners may readily adopt AI suggestions, this does not guarantee meaningful progress if guidance prioritises rapid solutions over sustainable strategies. This highlights the need for experimental research that examines not only the accuracy and timing of GenAI feedback but also its role relative to tutors' feedback in supporting deeper and more equitable learner engagement. Thus, in this study, we examine both tutor- and GenAI-generated feedback, drawing on their respective strengths to assess how each supports students' SRLs engagement in an authentic online learning setting.

Tutor and GenAI for educational feedback: a comparative evaluation

Research has demonstrated that advancements in LLMs have significantly enhanced AI's ability to provide educational written feedback, particularly in assessing student writing (Escalante et al., 2023; Meyer et al., 2024; Weber et al., 2024). LLMs operate advanced neural architectures to generate human-like text in response to natural language prompts. By predicting the most probable continuation of a given input based

on patterns learned from vast textual corpora, LLMs are able to produce contextually relevant and adaptive responses (Kosmyna et al., 2025).

Nevertheless, we have little empirical evidence about the performance of AI in creating SRL feedback in systematic and formative guidance compared to tutors. Addressing this gap is important because it can transform traditional feedback practices in online learning contexts and alleviate the tutor workload, opening the door to more effective SRLs interventions. To date, to the best of our knowledge, only a few studies, such as those by Dai et al. (2023), have explored the feedback capabilities of GenAI models, comparing the feedback from GPT-3.5 with that from tutors based on Hattie and Timperley's (2007) framework for effective feedback. A subsequent study by Dai et al. (2024) indicated that GPT-4 outperformed both GPT-3.5 and tutor feedback, with respect to feedback generation quality, such as readability. e. Specifically, they investigated the written feedback generated by tutors and AI across five assessment aspects (Goal, Topic, Benefit, Novelty, Clarity) and seven theoretical components of feedback (feeding up, feeding back, feeding forward, task, process, self-regulation, and self). Their findings suggested that the latest AI model surpassed tutor-generated content in the areas examined.

Additionally, Nazaretsky et al. (2024) investigated student perceptions of feedback generated by both tutors and GenAI, based on task-centred responses. Their examination utilised a framework encompassing four key dimensions: objectivity, usefulness, genuineness, and the credibility of the feedback provider. In their study, GenAI feedback was generated using GPT-4, informed by student responses and task descriptions. Conducted within authentic university course settings, their findings revealed that under blind review conditions, students rated GenAI feedback more favourably. However, once the identity of the feedback provider was disclosed, perceptions of GenAI feedback dropped, while tutor feedback received more positive ratings.

While these studies offer valuable insights into the comparison between AI-generated feedback and feedback written by tutors, it is necessary to study the effectiveness of such GenAI feedback interventions on various student learning outcomes in authentic-educational scenarios. Therefore, this study aimed to evaluate the practical deployment of data-driven GenAI-generated written feedback as an instrument to enhance SRLs feedback intervention in a higher education online learning context.

Methodology

Research context

Ethics approval for this study was granted by University College London (UCL) Institute of Education Ethics committee. The research context is a nine-week core module on introductory Statistics designed for higher education students. The module's objective is to cultivate students' comprehension of the role and application of statistics in academic and professional contexts. Over the course of nine weeks, students embarked on a multifaceted learning trajectory. Asynchronous online learning, which includes slide presentations, discussion forums, core and supplemental readings, and interactive workshops, facilitates weekly engagement.

Participants

The research participants comprise 94 higher education students enrolled in the Basic Statistics core module at a university in Türkiye during the second term of the 2023–2024 academic year and the first term of the 2024–2025 academic year. All students received detailed information about the research via an information sheet and were asked to provide informed consent, including demographic data (i.e., age, gender, GPA), at the commencement of the course. Participation was entirely voluntary. Invitations to participate were disseminated via email and a MOODLE course announcement.

Data collection

Data were collected from the LMS at three distinct stages, following a robust theoretical framework as presented below.

The framework for data collection to assess SRLs

Given that this research transforms a rich repository of raw log data generated by users in online learning environments into granular data for the assessment of SRLs by Ye and Pennisi's (2022) framework, it addresses the standard critique of LA research that the processing of raw data is often devoid of a robust theoretical foundation. Table 1 presents the framework for modelling log data in accordance with SRL theory and the online SRLs scale.

To convert raw log data into a format suitable for analysis, we employed a data mining process that involved extracting events relevant to the SRLs framework and normalising the dataset. The data were also standardised using Python scripts with the scikit-learn library, applying z-score standardisation to enable statistical analysis. All subsequent analyses were conducted on this standardised dataset. As shown in Table 1, the experiment was structured into three distinct stages. The first five weeks (Phase 0) focused on monitoring students' SRLs performance to gain insight into their initial status and to establish the pre-SRLs scores. Weeks 5–7 (Phase 1) constituted the intervention stage, during which two feedback interventions were offered. The final stage (Phase 2) involved final monitoring of students' SRLs once again to capture shifts in their engagement performance and to establish the post-SRLs scores. The feedback schedule was purposely designed based on the foundational research of Butler and Winne (1995). This research suggests that delayed feedback is particularly beneficial for tasks that require deep reflection and the application of learning strategies (Shute, 2008; Van Der Kleij et al., 2015), like SRL (Butler & Winne, 1995). Building on the foundational insights of Butler and Winne (1995), subsequent research has employed delayed feedback strategies in SRL contexts, yielding positive outcomes. For example, Lim et al. (2021) and Chen (2024) showed that delayed feedback can foster learners' engagement in self-regulatory processes. By allowing learners time to reflect before receiving feedback, they gain a better insight into their progress and can more effectively incorporate new strategies (Butler & Winne, 1995). Accordingly, this study offers delayed feedback as a mode to promote the SRLs.

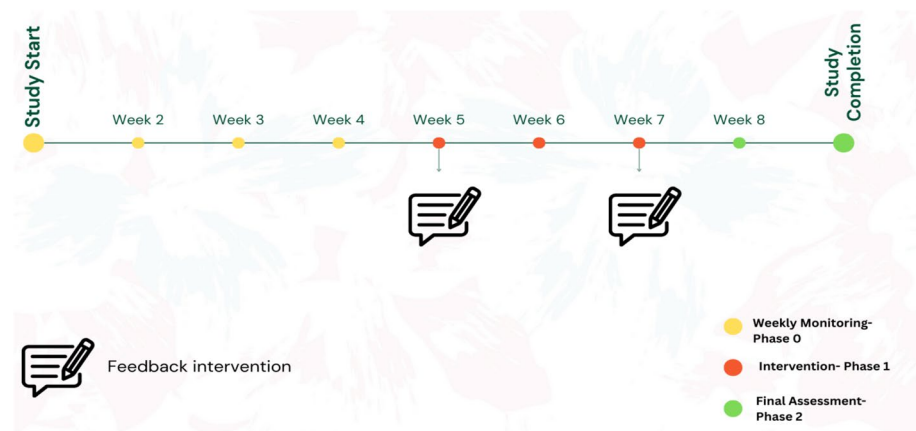


Fig. 1 Scheduled feedback interventions

Intervention design

Guided by the findings of Research Question 1, participants' SRLs proficiency was identified, and the intervention design was facilitated as summarised below in Fig. 1. Using a quasi-experimental design, participants were randomly assigned to either the control group (tutor-generated feedback) or the experimental group (GenAI feedback) from an existing list of participants. Participants in the experimental group were not informed about the source of feedback until the conclusion of the study, as explained in the GenAI-feedback Intervention section. Both groups engaged with the same content and materials, but received written feedback via email from different providers. Prior to the intervention, we verified that all participants could successfully receive messages in their inboxes. In instances where delivery issues occurred, participants were asked to provide an alternative email address to ensure the successful delivery of feedback. Consequently, all participants consistently received the feedback. On the tutor side, the same tutor was responsible for the entire course and was informed about the study's intervention schedule, including the existence of the GenAI-based feedback intervention.

Tutor-feedback intervention

Studies on tutor-generated feedback often emphasise the challenges associated with the level of expertise required to formulate effective and meaningful feedback (Banihashem et al., 2025). To mitigate this challenge, a practical tool, ReflexED, was developed in this study to support the provision of tutor-generated feedback. This tool displays each student's standardised SRLs proxy scores, cluster information based on the result of RQ1, and comprehensive insights into the SRLs pillars used in the study. Additionally, the (student_first_name) value associated with the student's ID is automatically retrieved from the database while the feedback is sent. Tutors, however, only have access to the student's ID, cluster, and standardised SRLs proxy scores, without visibility of the student's name. For each proxy, the "i" icon on the screen provides a tooltip containing the associated behavioural variables and their calculation methods. This feature was designed to assist tutors in interpreting the data and crafting tailored feedback messages, regardless of their familiarity

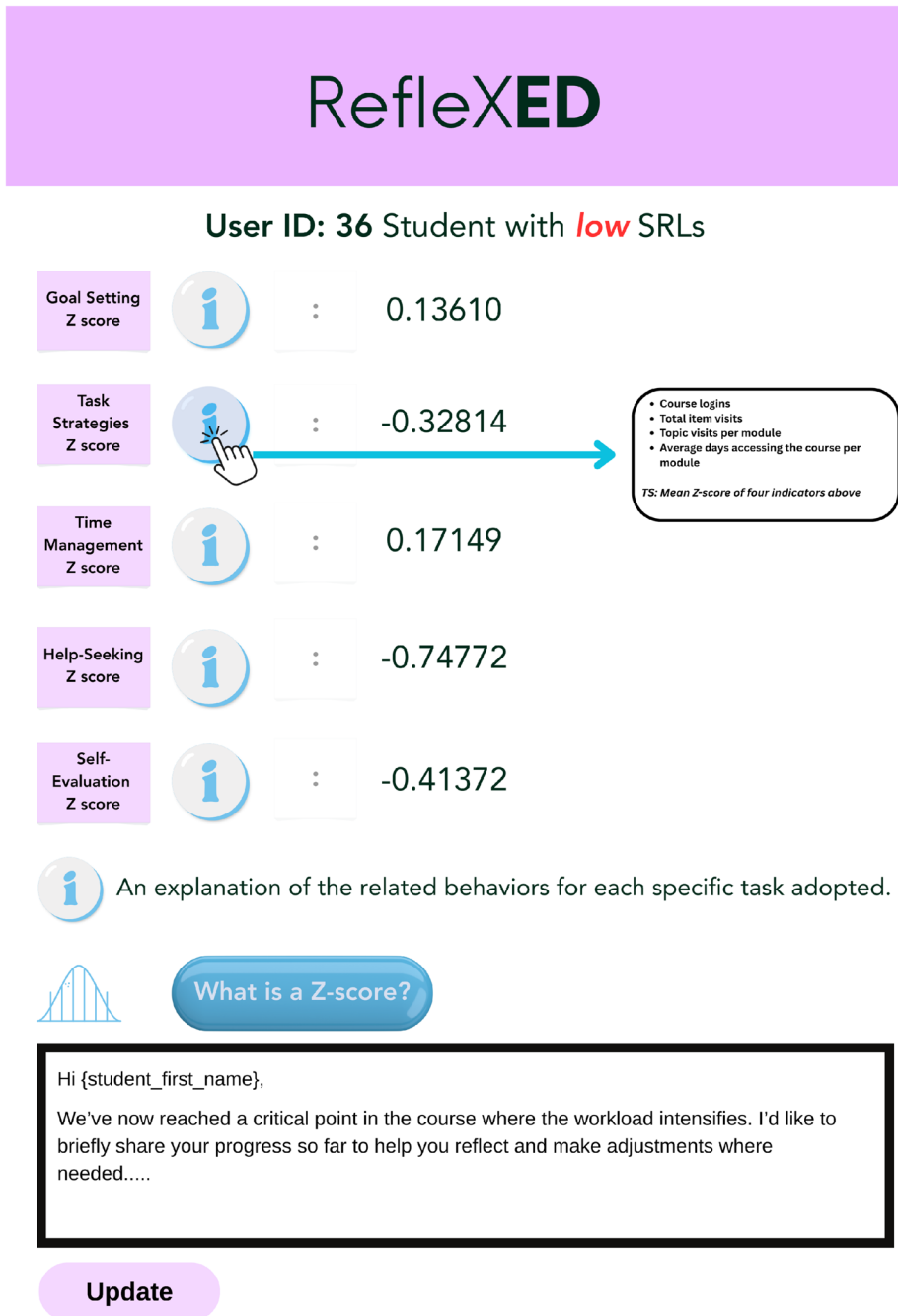


Fig. 2 Feedback creation screen of tutor. *This figure offers an illustrative representation of RefleXED, outlining its key functionalities

with statistical terms. In this study, however, the tutor was already familiar with statistical terms and had the necessary background to interpret the explanations and calculations for each pillar. Figure 2 illustrates the dedicated feedback creation interface used by the tutor to prepare personalised feedback for the control group. Once the tutor composed the feedback messages, they were stored in the database until the intervention date, allowing for revisions if needed. All students, regardless of their group, received feedback in clear, non-technical language, as shown in Appendix 1.

GenAI-feedback intervention

To create AI-generated feedback messages that support the development of SRLs, we utilised the ChatGPT-4 language model through the OpenAI API, guided by a specifically formulated set of prompts via a bespoke practical tool developed by the research team. Initially, the model was provided with detailed information regarding SRLs, including the skills adopted and the cluster groupings identified in the research with prompt engineering. This input enabled the model to tailor its feedback to each student's specific needs, based on their performance.

We tested the model's stability before the intervention across various performance clusters in a controlled ground. Finally, a structured prompt below was developed to guide the model in producing personalised feedback, incorporating each student's proxy Z-scores across the SRLs:

[Based on the information provided, generate a feedback message for {student_id} with {student_id_cluster} SRLs cluster, considering proxy Z-scores for Goal Setting {student_id_GS_score}, Task Strategy {student_id_TS_score}, Time Management {student_id_TM_score}, Help-Seeking {student_id_HS_score}, and Self-Evaluation {student_id_SE_score}. The feedback should include suggestions regarding the behavioural variables adopted in this study "(as detailed in Table 1)".]

Each item class in the structured prompt (e.g., {student_id_GS_score} and others) was connected to the individual student's Z-scores from the database, which personalises the feedback. The placeholders {student_id} and {student_id_cluster} are static and represent the student's unique identifier and cluster, respectively. In addition, the GenAI-feedback prompt included all the behavioural indicators listed in Table 1, which were referenced in the prompt as "as detailed in Table 1." These indicators were incorporated verbatim into the prompt text. They were a fixed component of the prompt template and appeared in every request sent to the LLM.

Given the two distinct feedback interventions in this study, the GenAI prompt for the second feedback intervention was designed to remain structurally consistent with the prompt used in the first intervention, differing only in the updated scores, which were provided in a timely manner. Prior to the second intervention, which followed the second round of behavioural data collection, we calculated the standardised score differences between each participant's first and second scores. For example, if a learner's initial TM score was 1.5 and later decreased to 1.2, the resulting change was calculated as -0.3 . This approach ensured that the prompt structure remained basically equivalent across both interventions, with only minimal adjustments to reflect score changes. Such consistency also enabled a systematic comparison of feedback quality across the two sources (RQ2).

Through these procedures, unlike traditional systems that apply rigorous, pre-fixed rules, our model directly embeds student-specific SRLs data into the prompt and dynamically processes it. By doing so, it moved beyond static, one-size-fits-all outputs, producing personalised (Dai et al., 2023; Jansen et al., 2024), context-sensitive feedback tailored to each learner's profile. To further examine adaptability, we implemented two distinct prompting strategies that leveraged the evolving SRLs proxy scores, enabling the model to adjust feedback dynamically to students' progress. This design emphasises a fundamental departure from traditional feedback systems: instead of reproducing fixed instructions with a narrow evaluative scale, our model provides feedback that is both dynamically data-driven and responsive to the learner's evolving needs. In doing so, we aimed to underscore students' evolving strengths (through positive feedback) and weaknesses (through negative feedback) across diverse SRLs proxies.

Consequently, since AI-generated content can raise ethical concerns, including potential bias, inappropriate language, or misleading evidence (Kumar et al., 2025), all GenAI feedback was carefully monitored. After each message was generated, the first author conducted a quality and ethics review before releasing it to students. No edits were deemed necessary during this intervention. Importantly, students in the GenAI-feedback group were not informed that their feedback had been generated by AI, which helped prevent a negative attitude toward the intervention. Upon completion of the experiment, however, these students were debriefed and informed about their condition. An example of the feedback messages sent is provided in Appendix 1.

Research methods

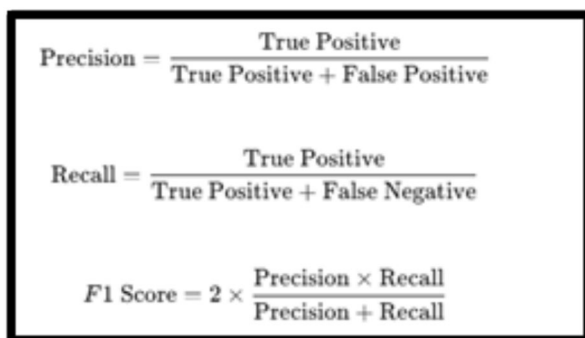
Identifying and categorising students' SRL profiles (RQ1)

To establish a baseline understanding of students' current SRLs status, we initially calculated their SRLs by analysing their interactions with the LMS. This allowed us to assess the varying levels of SRLs among students and identify their existing SRL profiles. Participants were categorised based on their performance in SRLs proxies before the first feedback intervention (Week 5). K-means cluster analysis was then used to group students exhibiting similar learning behaviours. This analysis focused on trace data identified as the pre-feedback point (Phase 0) in Fig. 1. The K-means clustering approach, widely utilised in similar studies (Du et al., 2021), enables the formation of groups without prior knowledge of participant cluster membership. The K-means algorithm effectively creates clusters based on shared learning patterns by calculating the distances between data points and cluster centroids. Python scripts utilising the scikit-learn library were employed for this analysis, specifically applying the K-means algorithm to perform clustering on the dataset. Clustering has become an essential tool in LA for processing trace data and addressing a wide range of research objectives (Bharara et al., 2018), specifically in the absence of standardised benchmarks for each criterion (Mojarad et al., 2018). The insights gained from this clustering process lay the groundwork for developing and customising feedback interventions tailored to learners' varying proficiency levels.

Measuring feedback readability and reliability (RQ2)

Since providing readable feedback is critical for helping students understand their learning trajectories (Black & Wiliam, 1998), this research specifically focused on readability

Fig. 3 Formulas of precision, recall and F1 metrics



$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

and reliability to evaluate the content quality of feedback interventions. The readability score was assessed by examining grammar, clarity, and coherence, as in the studies conducted by Dai et al., (2023, 2024). To assess the reliability of the feedback, we calculated the Precision, Recall, and F1 scores for each feedback message. These metrics are widely used to classify samples in multi-class cases (Chicco & Jurman, 2020). We invited two expert raters to assess all feedback from tutors and GenAI using a five-point scale. This scale ranged from 1 to 5, with 1 indicating incomprehensibility and 5 indicating entire fluency and coherence, as proposed by Dai et al. (2024). Two experts who were not directly involved in the research process only rated the feedback messages. The experts were provided with AI and tutor feedback in the same file, and the sources were not explicitly specified to ensure unbiased evaluations. To account for potential variations in readability assessments between the two expert raters, we calculated the average score given by the raters to establish the final readability score for each piece of feedback.

We posit that students exhibit varying qualifications in SRLs, which influence the nature of feedback interventions—either positive or negative. To evaluate the reliability of the generated feedback, we used LA-based standardised scores that led to the feedback interventions as the ground truth. In similar studies, tutor-generated feedback is often used as the baseline for comparison. However, we argue that tutor-generated feedback tends to be more generic due to the tutor's workload, whereas creating tailored feedback requires noteworthy resources. Therefore, we opted for LA-based standardised data as the ground truth to evaluate the reliability of feedback messages. To achieve this, we adapted five SRLs pillars for this study: Goal Setting, Task Strategies, Time Management, Help-seeking, and Self-Evaluation. Based on the polarity of each component in two standardised datasets that guided the feedback interventions, we calculated Precision, Recall, and F1 scores using the formulas shown in Fig. 3 (Reddy & Karthikeyan, 2022). In text classification, Precision refers to the proportion of predicted positives that are actually correct (Fränti & Mariescu-Istodor, 2023). Recall, on the other hand, calculates how effectively all true positive instances in the feedback are identified (Li et al., 2022). The F1 score is the harmonic mean of Precision and Recall, offering a balanced metric that handles any extreme values from either of the two metrics (Hicks et al., 2022).

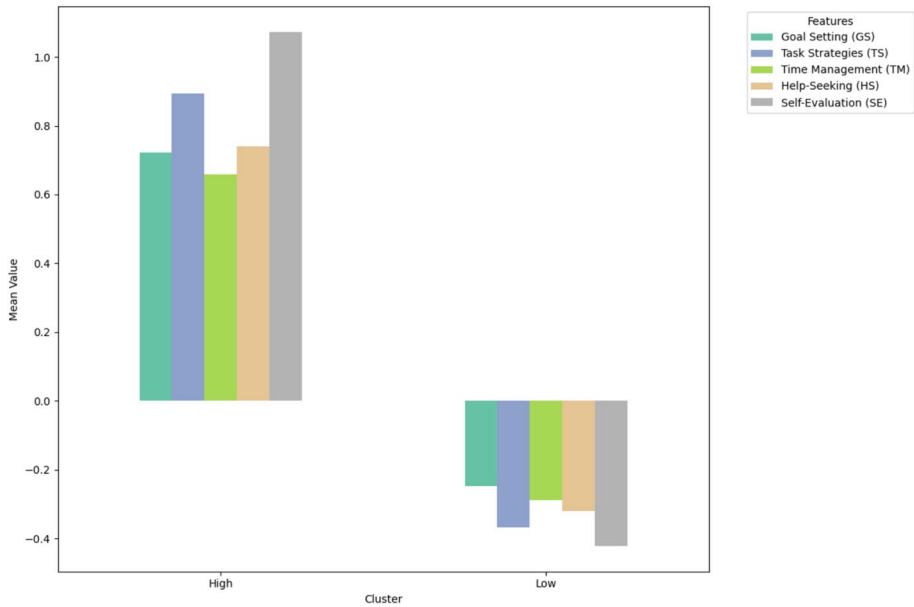


Fig. 4 Cluster information in the research

Quasi-experimental design to assess the impact of tutor and GenAI-generated feedback on students' SRLs (RQ3)

To address RQ3, a quasi-experimental design was employed, as the study involved distinct feedback groups. A pre–post design was adopted, whereby participants were assigned to different clusters and their SRLs status was measured at multiple stages. The experimental group received feedback generated by GenAI, while the control group received feedback from the tutor. The initial dataset, collected before the first intervention, served as a pre-score measure of students' SRLs, while the post-score data, gathered from the first intervention through the final week, captured the impact of the feedback strategies on the improvement of SRLs. Through the use of Ye and Pennisi's (2022) SRLs framework, we calculated SRLs proxies to reflect the five distinct pillars of the theoretical framework based on the indicators suggested for each pillar. However, our course did not include Optional Self-Assessments (SAssCom) and the bonus quiz (BQCom) completion rates; thus, these two indicators were excluded from the computation. In addition to the five proxies (GS, TS, TM, HS, and SE), the overall SRLs score was computed as the mean of the standardised proxy scores and subsequently employed to examine differences in engagement among clusters. Further details about this score and its interpretation are provided in the Results and Discussion sections.

Table 2 Minimum (Min), median, maximum (Max), mean, and standard deviation (Std) of readability scores of the feedback provided by the tutor, and GenAI

Feedback	Min	Median	Max	Mean	Std
Tutor	1.00	2.00	4.00	2.4043	.8464
GenAI	2.00	4.50	5.00	4.4149	.5269

Table 3 Mann–Whitney U test results for readability scores among tutor and GenAI-based feedback

Competency	Source	N	Mean rank	Sum of ranks	U	Z	p
Readability	Tutor	94	50.70	4765.5	300.5	– 11.193	< .001
	GenAI	94	138.30	13000.5			

Results

Identifying and categorising students' SRL profiles (RQ1)

Figure 4 exhibits the results of K-means clustering prior to the initial feedback intervention. The clustering analysis revealed two distinct groups: 'Students with high or low SRLs,' representing SRLs proxies. First, the silhouette score was calculated by the researcher to validate the optimal number of clusters. The silhouette coefficient score ($SC=0.5774$) indicated that the data fit well into two clusters. According to Kaufman and Rousseeuw (2009), a silhouette score of 0.5 or above suggests a reliable clustering solution. Based on SRLs behaviours, 52 participants demonstrated high SRLs proficiency, while 42 exhibited low SRLs proficiency. To ensure methodological consistency, the same analysis and clustering procedures were employed for both cohorts. The cluster labels identified in the first cohort were maintained, and the same labelling approach was applied to the second cohort. The final cluster labelling was summarised in one visual representation for clarity. Participants were randomly allocated into control and treatment groups based on the existing cluster results, which included 26 high-SRLs and 21 low-SRLs across both groups. This assignment resulted in four distinct conditions: AI-low, AI-high, Tutor-low, and Tutor-high.

Assessing the readability and reliability of feedback in relation to student SRLs (RQ2)

Table 2 presents the readability scores between GenAI and tutor feedback. Table 3 shows the statistical difference in readability between the tutor and GenAI. We can see that the feedback generated by GenAI was significantly more readable than tutor-generated feedback.

Figure 5 shows the distribution of readability scores for feedback given by both the tutor and GenAI. The boxplots illustrating these scores were produced employing the *seaborn* library in Python, allowing a comprehensive visualisation of the data distribution. The data shows that most readability scores for GenAI feedback were rated

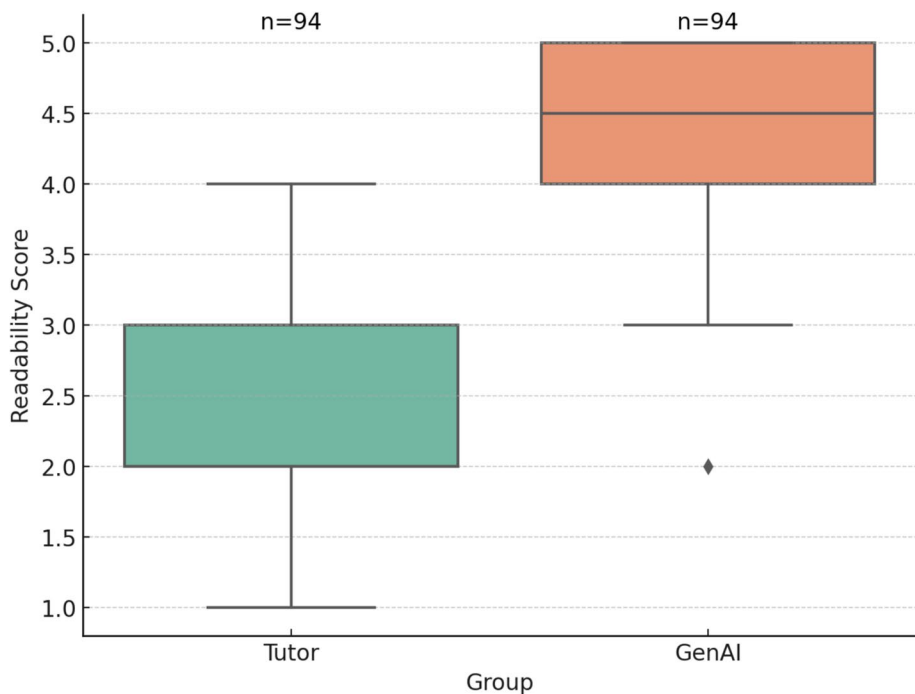


Fig. 5 Readability Scores of Feedback

between 4.0 and 5.0, while most scores for tutor feedback were below 3.0, with a higher standard deviation than GenAI feedback. The tutor's feedback predominantly consisted of simple expressions evaluating students' performance. For example: "I have noticed a drop in your course engagement. Since the next modules depend on previous ones, you should address this low engagement immediately. Please do not forget to stay in touch with me and your course mates if you need any help." Similarly, feedback such as, "It is good to see you are performing well in the course. Please continue to submit assignments on time and attend the course sessions as you have been," supports the argument in this research that tutor workload may lead to more generic feedback due to the "bandwidth problem" (Wiley, 2006). In contrast, feedback generated by GenAI typically provided a summary of students' performance using five distinct SRLs proxies, including an explanation of how participants could improve or maintain their status in SRLs. Table 3 shows that the Mann–Whitney U test revealed a significant difference in readability, favouring GenAI ($U = 300.5, p < 0.01$).

The results regarding the reliability of feedback are presented in Table 4. In terms of ground truth feedback polarity, specifically, standardised LA-based data that resulted in feedback interventions, GenAI more effectively addressed all five assessment aspects in every piece of feedback, unlike the tutor. Tutor-generated feedback frequently missed comments on some of these five aspects. Notably, 27 pieces of positive feedback overlooked comments on the help-seeking aspect, and 46 positive feedback did not address self-evaluation aspect. Accordingly, over half of the students did not receive comments on SRLs

Table 4 Polarity of feedback based on LA data and feedback by tutor and GenAI performance in five pillars of SRLs in the study, in terms of precision, recall and F1 reliability scores

Pillars	Polarity	Quantity				Precision		Recall		F1	
		LA-based data		Tutor	GenAI	Tutor	GenAI	Tutor	GenAI	Tutor	GenAI
GS	Positive	73	nT=38 nAI=35	21	29	0.50	0.53	0.63	0.85	0.56	0.65
	Negative	115	nT=57 nAI=58	20	25	0.48	0.50	0.62	0.83	0.54	0.62
TS	Positive	96	nT=47 nAI=49	23	36	0.65	0.73	0.62	0.78	0.63	0.75
	Negative	92	nT=48 nAI=44	28	31	0.70	0.70	0.66	0.75	0.68	0.73
TM	Positive	115	nT=61 nAI=54	20	38	0.80	0.71	0.68	0.79	0.74	0.75
	Negative	73	nT=34 nAI=39	18	23	0.78	0.60	0.66	0.69	0.72	0.64
HS	Positive	82	nT=40 nAI=42	13	23	0.61	0.65	0.68	0.58	0.65	0.62
	Negative	106	nT=55 nAI=51	15	33	0.65	0.73	0.71	0.67	0.68	0.70
SE	Positive	93	nT=48 nAI=45	2	28	0.66	0.59	0.13	0.70	0.22	0.64
	Negative	95	nT=47 nAI=48	13	26	0.92	0.57	0.5	0.68	0.65	0.62
Total		940									

components from the feedback they received in the tutor group. Compared to the tutor, GenAI demonstrated better performance in addressing all five aspects of SRLs feedback.

The Precision, Recall, and F1 scores, reported in Table 4, were calculated by evaluating whether the feedback polarity (positive or negative) generated by the two sources aligned with the ground truth labels provided in the LA datasets.

The values presented in Table 4 clearly indicate that the ground truth, LA-based data, primarily indicates negative feedback, meaning a need for improvement across various groups. The table shows only valid feedback polarity based on LA data, excluding null and incorrectly addressed responses. Nonetheless, incorrectly addressed feedback, such as "false negatives" and "false positives", was included in the calculations for Precision, Recall, and F1 scores. The data shows that GenAI provided 138 instances of negative feedback out of 240 instances, while the tutor issued 94 instances out of 241.

Although several instances are recorded as null for the tutor, it appears that both the Tutor and GenAI have similar Precision values in many cases, suggesting that both sources generally provide accurate feedback. However, there are instances where one source outperforms the other. For example, GenAI shows higher Recall in several cases of negative feedback (specifically in the GS, TS, TM, and SE pillars), indicating that GenAI is more effective at identifying areas needing improvement. Conversely, the tutor exhibits relatively

high Recall in the HS pillar, where its performance is comparable to or slightly better than that of GenAI.

GenAI generally achieves higher F1 scores in most instances, especially when assessing positive feedback. This indicates that GenAI offers a more balanced approach, effectively identifying situations that require both positive and negative feedback. Notably, in the SE positive feedback category, GenAI's F1 score is remarkably higher than that of the tutor, with scores of 0.64 compared to 0.22.

Assessing the impact of tutor and GenAI-generated feedback on students' SRLs (RQ3)

The results of the normality tests for four distinct conditions of pre- and post-scores indicate that the research dataset is suitable for non-parametric tests, particularly due to the distribution of the AI-low condition data ($p < 0.05$). Therefore, non-parametric tests were used to examine statistical differences between conditions.

This research involved four distinct conditions, and we compared participants at the same level of proficiency in SRLs across different groups. Specifically, we compared the low SRLs GenAI condition with the low SRLs tutor condition and the high SRLs GenAI condition with the high SRLs tutor condition. According to the pre-scores, no significant differences were found between the same-level conditions, indicating that the conditions in the different groups were comparable prior to the intervention, as shown in Table 5. For post-score, the results of the Mann–Whitney U test revealed significant differences in the post-Goal Setting mean rank scores for participants in the low condition. Specifically, there were significant differences between the Tutor-low control condition (16.83) and the AI-low experimental condition (26.17), with the AI-based low cluster performing better ($U = 122.500$, $p = 0.012$, $r = 0.388$). This suggests that the AI-based approach had a more significant positive effect on the low cluster in Goal Setting, with a moderate effect size as indicated by the Rank-Biserial Correlation ($r = 0.388$) (Martin & Martinez, 2023). Furthermore, the analysis of the composite SRLs score revealed a similar pattern. This score, calculated as the mean of the standardised SRL proxies in this study, represents a composite construct that captures the differences in students' overall self-regulatory engagement and should be interpreted at the level of the construct score, given the nature of our mean-rank-based analysis. Specifically, when the composite score is analysed using the Mann–Whitney U test, all observations across both groups are converted into ranks (from low to high across the combined sample), the ranks are summed within each group, and the U statistic is then calculated based on these rank sums rather than the original values. Participants in the low-SRLs condition who received GenAI feedback showed significantly higher post-intervention composite SRLs mean rank scores compared to those in the Tutor condition ($U = 133.00$, $p = .028$, $r = .340$). This result aligns with the findings on Goal Setting, suggesting better performance for the GenAI-based low cluster. No significant differences were found in other skills among the low SRLs condition post-scores (TS: $U = 154.0$, $p = .094$; TM: $U = 175.5$, $p = .256$; HS: $U = 152.5$, $p = .083$; SE: $U = 145.0$, $p = .057$). Notably, the Tutor-low group exhibited a systematic decrease in mean rank of SRLs score relative to their pre-intervention baseline, whereas the GenAI-low group showed increases across all SRL pillars. On the other hand, for the high condition, no significant differences were observed before or after the intervention, consistent with our initial assumption. High-cluster students in the Tutor group had a mean rank of SRLs score of 27.50 at the pre-phase and 27.12 at the post-phase, while the GenAI-high group scored a mean rank of

Table 5 Comparison of SRLs pillars and SRLs pre-post scores between same-level clusters

Pillar	Stage	genai-high ($n=26$, mean rank)	Tutor-high ($n=26$, mean rank)	Sig. (high)	GenAI-low ($n=21$, mean rank)	Tutor-low ($n=21$, mean rank)	Sig. (low)
GS	Pre	26.27	26.73	$U=332.0, p=.913$	23.40	19.60	$U=180.5, p=.314$
	Post	24.02	28.98	$U=273.5, p=.237$	26.17	16.83	$U=122.5, p=.012^*, r=.388$
TS	Pre	25.46	27.54	$U=311.0, p=.621$	20.07	22.93	$U=190.5, p=.450$
	Post	25.08	27.92	$U=301.0, p=.498$	24.67	18.33	$U=154.0, p=.094$
TM	Pre	27.00	26.00	$U=325.0, p=.812$	20.64	22.36	$U=202.5, p=.650$
	Post	29.19	23.81	$U=268.0, p=.200$	23.64	19.36	$U=175.5, p=.256$
HS	Pre	27.02	25.98	$U=324.5, p=.804$	21.50	21.50	$U=220.5, p=1.000$
	Post	28.38	24.62	$U=289.0, p=.370$	24.74	18.26	$U=152.5, p=.083$
SE	Pre	25.06	27.94	$U=300.5, p=.492$	21.90	21.10	$U=212.0, p=.830$
	Post	25.00	28.00	$U=299.0, p=.475$	25.10	17.90	$U=145.0, p=.057$
SRLs	Pre	25.50	27.50	$U=312.0, p=.634$	21.10	21.89	$U=212.0, p=.831$
	Post	25.88	27.12	$U=322.0, p=.770$	25.67	17.33	$U=133.0, p=.028^*, r=.340$

25.50 at the pre-phase and 25.88 at the post-phase. Accordingly, no statistically significant differences were found in SRLs proxies at either the pre- or post-stage for high-cluster students. Table 5 reports the SRLs pillars and overall mean rank scores across conditions.

Discussion

Identifying and categorising student SRL profiles

This study identified distinct SRLs behaviours among students by analysing LMS log data using the Ye and Pennisi (2022) framework. The findings, in line with our hypothesis, reveal that students exhibit varying levels of SRLs proficiency, categorised as high and low. These distinct SRL profiles are aligned with Pintrich's SRL model (Pintrich, 2000), which suggests that disparities in how learners regulate their cognition, motivation, and behaviour, shaped by both contextual and individual factors, can lead to higher or lower levels of profiles. High-regulated students actively employ strategic learning behaviours, such as engaging with course materials promptly after receiving external support (Chen & Li, 2021). In contrast, low-SRLs students demonstrate limited engagement and self-regulatory control (Schunk & Ertmer, 2000). In online learning environments, where students are expected to take an active role in their learning process, students with low SRLs proficiency face significant disadvantages compared to their high-SRLs peers, who effectively employ SRLs. Therefore, identifying, supporting, and monitoring different SRLs clusters was crucial for providing appropriate support and ensuring equitable learning opportunities, especially for low SRLs who struggle to navigate online learning environments effectively, potentially hindering their academic success (Cleary et al., 2021). Instead of assuming that all students share identical learning needs and using a uniform intervention, this study utilised clustering to identify distinct SRL profiles. This approach allowed for a more nuanced understanding of students' SRLs and learning behaviours, ensuring that support strategies can be aligned with their specific needs. By recognising the diverse profile of SRLs' proficiency, clustering serves this research purpose in creating feedback tailored to individual learning patterns. Our clustering results, which categorised students into high and low SRL proficiency groups, align with previous research such as Yot-Domínguez and Marcelo (2017), who also identified two distinct SRL profiles operating a wide range of clustering methods, including K-means clustering. However, Yot-Domínguez and Marcelo (2017) collected their data via self-report to identify students' regulation level. Despite this difference, both studies highlight the presence of two distinct SRL profiles, suggesting that SRL proficiency can actually be categorised into high and low groups using the same clustering techniques.

In addition, our clustering results align with the findings of Ye and Pennisi (2022). In their study with 65 students, they used K-means clustering to trace data and identified both a two-profile (low and high) and a three-profile (low, moderate, and high) SRL, each reflecting unique learner characteristics. They observed that the moderate cluster frequently displayed characteristics resembling both the low and high clusters in various SRL variables. This suggests that a two-profile solution could also be valid. In comparison, our study involved 94 participants and employed distinct methodological approaches. Specifically, Ye and Pennisi (2022) clustered students based on individual SRL variables within each pillar. In contrast, our study integrates these variables into broader categories: GS, TS, and others, to reduce tutors' cognitive load and ensure a more manageable, context-sensitive

feedback process, without overwhelming the tutor with more than 20 separate indicators. This method streamlines feedback writing, enabling the tutor to focus on the actual learning scenario rather than navigating an overly complex indicator structure. These methodological differences may account for slight variations in clustering results. Nevertheless, in agreement with Ye and Pennisi (2022), our findings suggest that students can indeed be categorised into two distinct SRL profiles.

As a result, two emergent clusters in our population were considered stable and were taken forward for study in terms of the impact of feedback intervention on them.

Feedback readability and reliability

The results of RQ2 showed that the GenAI model generated more readable feedback than the tutor. The statistical comparison revealed a significant difference between the two sources of feedback, favouring GenAI. This is not surprising, as the ability of LLMs to produce high-quality responses across various contexts has been shown in the literature (Dai et al., 2024; Meyer et al., 2024). As argued in this research, tutors often struggle to deliver high-quality feedback that aligns with students' expectations (Nazaretsky et al., 2024; Van Der Kleij et al., 2015). This can result in generic and disagreeing feedback (Dai et al., 2024; Nazaretsky et al., 2024), which may demotivate students and restrict their understanding (Ackerman-Barger et al., 2016). Additionally, given that contemporary LA-based systems for providing feedback still rely heavily on tutors (Pardo et al., 2019), based on our findings, these limitations could be mitigated by integrating LLMs, empowered by LA, to assist tutors in providing feedback with consistent readability, thereby improving students' comprehension.

As for reliability, we used our LA-based datasets as the ground truth. Unlike Dai et al. (2024), who relied on tutor feedback as the ground truth despite its limitations, we argue that LA datasets provide a more robust and objective baseline for evaluation. To reduce the expertise barrier in producing feedback for the control group, the tutor was supported by the RefleXED tool, which dynamically displayed students' SRL performance prior to each intervention. This tool ensured a fairer comparison between the feedback sources. However, we acknowledge that if the tutor had been responsible for fewer students than GenAI, the outcomes might have differed. Given that this study represents one of the initial attempts to examine GenAI's capacity to generate feedback and provide longitudinal support to students, it was essential to compare its performance against that of a tutor supported by the tool under an authentic and comparable online learning condition. Accordingly, we examined the performance of both sources under comparable conditions.

In this context, calculating reliability between tutor- and GenAI-generated feedback in our implementation is particularly noteworthy, as even with such support, tutors face persistent constraints (e.g., time and resources) that make it challenging to address students' evolving needs comprehensively. Comparing the two sources, therefore, offers insights into the complementary and practical role that GenAI can play within a longitudinal feedback support model.

It is also noteworthy to emphasise that our GenAI results were not driven by random prompt engineering choices, but by systematically altering the learner's evolving score embedded within a designated prompting structure. While different prompting strategies can indeed influence outcomes, our design intentionally controlled for this factor by holding the prompt template constant and varying only the evolving learner scores (i.e., second

scores versus the first scores). This approach ensured that feedback reliability was evaluated within a consistent framework.

As shown in Table 4, in terms of precision, recall, and F1 scores, GenAI generally outperformed tutor-generated feedback, particularly in recall and precision. It tends to show higher precision, while recall values are sometimes closer to those of the tutor. In certain areas, such as TM, the tutor provides higher precision and F1 scores; however, GenAI feedback consistently yields more balanced results. With the caveat of the higher null values in tutor feedback compared to GenAI, these results suggest that GenAI is generally capable of generating accurate and consistent SRLs feedback based on analytics proxies, performing more effectively than the tutor in this online learning context.

This result also aligns with Banihashem et al.'s (2024) implication on the promising future use of AI in writing feedback. As a result, this finding suggests that the GenAI model can provide reliable insights into whether students performed well or poorly in terms of feedback alignment with the LA-based data. Similarly, our findings align with Dai et al. (2024), underscoring GenAI's potential to produce reliable feedback under conditions where tutors may be constrained by workload. Furthermore, these findings align with the hypothesis of this research, which stems from the resource and time limitations tutors face in delivering precise feedback.

Nonetheless, we argue that a proficient tutor, with adequate resources, can potentially still provide higher-quality, content-aware, and personalised feedback than any machine, tool, or program. However, this condition was not tested in this study. The primary aim of this research is to demonstrate AI's potential within an authentic online learning context, where tutors have limited time and a high workload.

Impact of feedback interventions on SRLs

The third research question explored shifts in SRLs during the study, specifically through engagement with the online learning environment. As shown in Table 5, after the first intervention, GenAI feedback had a notable impact on GS skills and overall SRLs performance among low-SRL students. This suggests that GenAI-driven feedback may help enhance the self-regulation of low-achieving learners. However, it is important to note that significant improvements were observed only in GS and the overall SRLs score, which represents a composite score and should be interpreted as a single construct, rather than representing its sub-dimensions, aligning with the non-parametric nature of our analysis, which compared the mean ranks of engagement scores. For other critical dimensions, including TS, TM, HS, and SE, although the low-SRL GenAI group showed an upward trend in mean ranks, no significant differences were found compared to the low-SRL tutor group. The improvement may be explained by the more readable and tailored feedback provided by GenAI, which better addressed students' goals and guided their progress. In contrast, low-SRLs students who received tutor feedback experienced a decline in their SRLs scores, as shown in Table 5. This drop may be due to the fact that students receiving generic feedback tend to show a decrease in performance related to their self-regulation, as Butler and Winne (1995) stated. They argue that students with a gap between their current and desired performance require more comprehensive external feedback, rather than focusing solely on conceptual understanding. Even when students are willing to engage with feedback (Kizilcec et al., 2017), research indicates they are more likely to do so when they perceive the feedback as personalised and reflective of their individual progress and status (Ackerman-Barger et al., 2016).

Additionally, as Steinbach et al. (2025) note, learners with initially low performance benefit most from targeted feedback and may struggle to remain engaged when feedback does not meet their evolving needs. This may be related to the type of data used to generate feedback. However, as noted in the Tutor-Feedback Intervention section, the tutor in this study possessed a solid technical and statistical background. Therefore, we do not attribute the observed findings to this factor when interpreting the results. Consequently, the results observed in the control group, which received feedback from the tutor, align with these discussions and support this study's arguments regarding the challenges of traditional feedback provision such as the generic nature of the feedback (Dai et al., 2024; Nazaretsky et al., 2024; Wiley, 2006), often stemming from tutor workload constraints (Meyer et al., 2024; Wiley, 2006). These findings also confirm prior research showing that low-SRL students are the most at risk of being negatively affected by such persistent challenges (Kizilcec et al., 2017). However, the significant differences observed in only one pillar, GS, and in the overall SRLs engagement score with moderate effects should be interpreted with caution. While there was an upward trend across all SRL pillars in the low-SRL group receiving GenAI feedback and a decrease in mean ranks for the tutor group within the same cluster, this does not necessarily indicate an overall superiority of GenAI feedback over tutor intervention. Instead, it highlights the potential to leverage GenAI's strengths in educational feedback interventions.

This finding not only sheds light on GenAI's practical potential but also resonates with the insights of Chine et al. (2022), who examined hybrid AI- and human-based interventions. They emphasised the complementary strengths of both sources: while AI offered data-driven recommendations, human tutors provided contextual and pedagogical insights, which resulted in substantial learning gains, particularly among lower-achieving students. Similarly, our results align with those of Ng et al. (2024), who reported statistically significant improvements in students' SRL skills, even though the effect sizes varied. However, a significant methodological difference should be noted. Whereas Ng et al. (2024) compared an AI-based tool with a pre-fixed, rule-based system, our study focused on the practical applicability of GenAI by directly comparing tutor-generated and GenAI-generated feedback in authentic educational settings. This distinction in research design may help explain the variation in effect sizes observed between the two studies.

On the other hand, students with high SRLs did not show significant differences in their SRL performance across the SRLs pillars and overall SRLs performance after the first feedback intervention. Examination of the mean engagement ranks for the high-SRL clusters indicated minimal variation. For instance, the GenAI-high group demonstrated an overall SRLs mean rank of 25.50 pre-intervention and 25.88 post-intervention. In contrast, the Tutor-high group had a pre-intervention mean rank of 27.50, which slightly decreased to 27.12 post-intervention. This is expected, as some scholars argue that highly self-regulated learners already engage in more effective learning behaviours (Van Alten et al., 2021; Verstege et al., 2019). This may also be due to the upper-performance threshold effect, where these students have limited space for further progress. Additionally, these students likely have high standardised values across SRL skills pillars, suggesting that the feedback they receive is more likely to reinforce their existing efforts. In other words, providing positive feedback to encourage continued use of their existing strategies might have led them to preserve *the status quo*.

Consequently, the results suggest that incorporating LA-powered GenAI support into online learning environments' feedback provision may enhance SRLs for the

lower-achieving group. Tutors can also benefit from this support by creating tailored interventions, such as an LA-driven, GenAI-informed, tutor-led feedback model. This approach could reduce tutors' workload while ensuring more personalised feedback for students. Given that GenAI did not demonstrate absolute superiority over tutors in our study, the potential value of a complementary relationship between the two warrants consideration, emphasising the distinct strengths each source can contribute. As we initially argued, considering the diversity of learners in the classroom, our model has the potential to foster engagement in certain learner characteristics, specifically by addressing the needs of lower-achieving students.

Limitations and future research

This research has several limitations, including participant awareness issues, as well as epistemological and cognitive considerations that should be acknowledged. First, while participants in the AI-feedback group are blinded to the feedback source until the study's completion, implicit expectations could still occur unconsciously based on the content of the feedback. For example, differences in the tone or style of the feedback might lead participants to form subconscious beliefs about its origin, even if they are unaware of it. These beliefs, in turn, could affect their reactions and how they interpret the feedback, potentially influencing the overall outcomes. Future research should therefore examine students' awareness of feedback sources (e.g., Nazaretsky et al., 2024) and whether such awareness modifies their responses. Similarly, the tutor's awareness of the intervention, could have influenced the quality of feedback in different ways. On one hand, it may have encouraged more intentional feedback; on the other hand, it may have affected the consistency of the feedback. To further explore this notion, the readability and reliability of feedback messages could be examined across different patterns, such as within the feedback corpus, among students with distinct SRL profiles, or particularly for those with very high or very low standardised engagement scores, which might uncover patterns in the tutor's feedback behaviours associated with their awareness of GenAI intervention and help explain the observed findings. In addition, a potential remedy would be to withhold information about GenAI's involvement until after all interventions are completed, ensuring that the tutor's feedback remains consistent with regular practice.

Another limitation lies in the use of LLMs as feedback generators. Despite its advanced text generation capabilities, the model's training data are prone to inaccuracies and biases (Kumar et al., 2025). Monitoring and refining generated feedback thus added to the researcher's workload (a dimension not counted as part of the tutor's workload in this study). Since our argument centres on existing tutor workload in providing feedback, future research should explicitly compare researcher and tutor workload. More significantly, while automation promises efficiency, adopting it solely on the basis of technical feasibility risks undermining the cognitive competencies that underpin adaptability and intellectual development (Cukurova, 2025; Lan & Zhou, 2025). Future studies should therefore aim to ensure that students continue to cultivate intrinsic abilities rather than becoming overly dependent on external feedback.

This study also relied on LMS log data and behavioural measures, which, although valid, carry epistemological limitations. LMS logs capture only observable actions (Bergdahl et al., 2024) such as clicks, time on task, or engagement patterns (Maloney

et al., 2022) and do not necessarily reflect learners' internal cognitive or metacognitive processes (Hadwin et al., 2007; Gasevic et al., 2017). For example, high engagement might signal effective SRL, but it could equally indicate hesitation or inefficient study habits. To provide a more comprehensive understanding of learning, qualitative data should therefore complement behavioural measures (Bergdahl et al., 2024).

Lastly, given our strategies of producing feedback and findings observed, future research should also explore the effects of GenAI interventions, especially for students with high SRL skills. This investigation should include different strategies than those we employed.

Conclusion

This study examined the effect of an LA-based GenAI and tutor-generated feedback on students' self-regulated learning skills in an online nine-week course within higher education contexts. We categorised students into low and high SRL skill levels according to their status. We assessed the readability and reliability of feedback from both sources, where GenAI outperformed tutors in terms of readability and providing more tailored feedback. Although there were no significant overall differences in SRL skills between all conditions, our results revealed that low-SRL students who received GenAI feedback showed noticeable improvements in a specific SRL component, goal setting, and in overall composite SRLs. These findings underscore the potential of incorporating LA-based GenAI-generated feedback interventions into online learning settings, particularly for students who struggle with self-regulation. Notably, these findings provide pathways for further investigation addressing the limitations of this study, rather than presenting definitive conclusions. The paper concludes that providing LA-informed feedback through GenAI may offer better support for low-SRL students' specific skills and, at scale, can also enhance their engagement performance in online learning contexts.

Appendix

Appendix 1 Feedback examples from GenAI and Tutor, See Table 6.

Table 6 Feedback examples from GenAI and tutor

Example of Feedback Messages
GenAI feedback message
Hi {StudentFirstname}, First of all, I would like to provide some view on the changes in your SRLs performance over the past few weeks. We've observed a positive shift in your goal setting and time management skills. However, you may have encountered some challenges with task strategies, help seeking, and self-assessment. Keep in mind that improving these areas can have a significant impact on your overall course success. To enhance your task strategies, increased participation in course sessions could be beneficial. Don't hesitate to ask more questions to both your instructors and peers to strengthen your help seeking skills. For self-assessment, consider revisiting the module after grades are announced to better reflect on your progress. I'm confident that you will not only achieve your goals but also enhance your self-regulation capabilities. Keep up the hard work – I believe in your potential. Best regards.
Tutor feedback message
Hi {StudentFirstname}, We've reached week5, marking the beginning of a more work-intensive period in the course and I've noticed a decline in your participation in key class activities, which are essential for your success in the upcoming weeks. I recommend reviewing the course materials and revisiting any topics where you don't feel as confident. Please feel free to share any points you're having difficulty with or our classmates. Additionally, don't forget that you can edit and resubmit your homework until the deadline. Keep up the great work. Warm regards.

Author contributions First Author: Formal analysis, investigation, writing, and development of the practical tool. Second and Third Authors: Supervision, review, and editing.

Data availability The data used to support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this study.

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References

- Ackerman-Barger, K., Valderama-Wallace, C., Latimore, D., & Drake, C. (2016). Stereotype threat susceptibility among minority health professions students. *Journal of Best Practices in Health Professions Diversity*, 9(2), 1232–1246.
- Afacan, A. G., & Gülbahar, G. Y. (2023). Learner autonomy in online learning: Development and validation of a scale. *Journal of Higher Education and Science*, 13(1), 116–125. <https://doi.org/10.5961/higheredusci.1152249>
- Alhazbi, S., Al-ali, A., Tabassum, A., Al-Ali, A., Al-Emadi, A., Khattab, T., & Hasan, M. A. (2024). Using learning analytics to measure self-regulated learning: A systematic review of empirical studies in higher education. *Journal of Computer Assisted Learning*, 40(4), 1658–1674. <https://doi.org/10.1111/jcal.12982>
- Banihashem, S. K., Gašević, D., & Noroozi, O. (2025). A critical review of using learning analytics for formative assessment: Progress, pitfalls and path forward. *Journal of Computer Assisted Learning*, 41(3), Article e70056. <https://doi.org/10.1111/jcal.70056>
- Banihashem, S. K., Kerman, N. T., Noroozi, O., Moon, J., & Drachsler, H. (2024). Feedback sources in essay writing: Peer-generated or AI-generated feedback? *International Journal of Educational Technology in Higher Education*, 21(1), Article 23. <https://doi.org/10.1186/s41239-024-00455-4>
- Barnard, L., Lan, W. Y., To, Y. M., Paton, V. O., & Lai, S.-L. (2009). Measuring self-regulation in online and blended learning environments. *The Internet and Higher Education*, 12(1), 1–6. <https://doi.org/10.1016/j.iheduc.2008.10.005>
- Barz, N., Benick, M., Dörrenbächer-Ulrich, L., & Perels, F. (2024). Fostering self-regulated learning using synchronous or asynchronous digital learning environments: A latent profile analysis of pre-service teachers' individual differences. *Frontiers in Education*, 9, Article 1445182. <https://doi.org/10.3389/educ.2024.1445182>
- Bergdahl, N., Bond, M., Sjöberg, J., Dougherty, M., & Oxley, E. (2024). Unpacking student engagement in higher education learning analytics: A systematic review. *International Journal of Educational Technology in Higher Education*, 21(1), Article 63. <https://doi.org/10.1186/s41239-024-00493-y>
- Bernacki, M. L. (2017). Examining the cyclical, loosely sequenced, and contingent features of self-regulated learning: Trace data and their analysis. *Handbook of self-regulation of learning and performance* (pp. 370–387). Routledge.
- Bharara, S., Sabitha, S., & Bansal, A. (2018). Application of learning analytics using clustering data mining for students' disposition analysis. *Education and Information Technologies*, 23(2), 957–984. <https://doi.org/10.1007/s10639-017-9645-7>
- Bjork, R. A., Dunlosky, J., & Kornell, N. (2013). Self-regulated learning: Beliefs, techniques, and illusions. *Annual Review of Psychology*, 64(1), 417–444. <https://doi.org/10.1146/annurev-psych-113011-143823>
- Black, P., & Wiliam, D. (1998). Assessment and classroom learning. *Assessment in Education: Principles, Policy & Practice*, 5(1), 7–74. <https://doi.org/10.1080/0969595980050102>
- Bozkurt, A. (2023). Generative artificial intelligence (AI) powered conversational educational agents: The inevitable paradigm shift. *Asian Journal of Distance Education*, 18(1), 198–204. <https://doi.org/10.5281/zenodo.7716416>
- Butler, D. L., & Winne, P. H. (1995). Feedback and self-regulated learning: A theoretical synthesis. *Review of Educational Research*, 65(3), 245–281.
- Chen, J. (2024). Effects of learning analytics-based feedback on students' self-regulated learning and academic achievement in a blended EFL course. *System*, 124, Article 103388. <https://doi.org/10.1016/j.system.2024.103388>
- Chen, K.-Z., & Li, S.-C. (2021). Sequential, typological, and academic dynamics of self-regulated learners: Learning analytics of an undergraduate chemistry online course. *Computers and Education: Artificial Intelligence*, 2, Article 100024. <https://doi.org/10.1016/j.caeai.2021.100024>
- Chicco, D., & Jurman, G. (2020). The advantages of the Matthews correlation coefficient (Mcc) over F1 score and accuracy in binary classification evaluation. *BMC Genomics*, 21(1), Article 6. <https://doi.org/10.1186/s12864-019-6413-7>
- Chine, D. R., Brentley, C., Thomas-Browne, C., Richey, J. E., Gul, A., Carvalho, P. F., Branstetter, L., & Koedinger, K. R. (2022). Educational equity through combined human-ai personalization: A propensity matching evaluation. In: M. M. Rodrigo, N. Matsuda, A. I. Cristea, & V. Dimitrova (Eds), *Artificial Intelligence in Education*. (Vol. 13355, pp.366–377) Springer

- Chiu, T. K. F., Xia, Q., Zhou, X., Chai, C. S., & Cheng, M. (2023). Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 4, Article 100118. <https://doi.org/10.1016/j.caeai.2022.100118>
- Cleary, T. J., Slemple, J., & Pawlo, E. R. (2021). Linking student self-regulated learning profiles to achievement and engagement in mathematics. *Psychology in the Schools*, 58(3), 443–457. <https://doi.org/10.1002/pits.22456>
- Cukurova, M. (2025). The interplay of learning, analytics and artificial intelligence in education: A vision for hybrid intelligence. *British Journal of Educational Technology*, 56(2), 469–488. <https://doi.org/10.1111/bjjet.13514>
- Dai, W., Lin, J., Jin, H., Li, T., Tsai, Y.-S., Gašević, D., & Chen, G. (2023). Can large language models provide feedback to students? A case study on chatgpt. *IEEE International Conference on Advanced Learning Technologies (ICALT)*, 2023, 323–325. <https://doi.org/10.1109/ICALT58122.2023.00100>
- Dai, W., Tsai, Y.-S., Lin, J., Aldino, A., Jin, H., Li, T., Gašević, D., & Chen, G. (2024). Assessing the proficiency of large language models in automatic feedback generation: An evaluation study. *Computers and Education: Artificial Intelligence*, 7, Article 100299. <https://doi.org/10.1016/j.caeai.2024.100299>
- Du, J., Hew, K. F., & Liu, L. (2023). What can online traces tell us about students' self-regulated learning? A systematic review of online trace data analysis. *Computers & Education*, 201, Article 104828. <https://doi.org/10.1016/j.compedu.2023.104828>
- Du, X., Yang, J., Shelton, B. E., Hung, J.-L., & Zhang, M. (2021). A systematic meta-review and analysis of learning analytics research. *Behaviour & Information Technology*, 40(1), 49–62. <https://doi.org/10.1080/0144929X.2019.1669712>
- Edisherashvili, N., Saks, K., Pedaste, M., & Leijen, Ä. (2022). Supporting self-regulated learning in distance learning contexts at higher education level: Systematic literature review. *Frontiers in Psychology*, 12, Article 792422. <https://doi.org/10.3389/fpsyg.2021.792422>
- Escalante, J., Pack, A., & Barrett, A. (2023). AI-generated feedback on writing: Insights into efficacy and ENL student preference. *International Journal of Educational Technology in Higher Education*, 20(1), Article 57. <https://doi.org/10.1186/s41239-023-00425-2>
- Fan, Y., Matcha, W., Uzir, N. A., Wang, Q., & Gašević, D. (2021). Learning analytics to reveal links between learning design and self-regulated learning. *International Journal of Artificial Intelligence in Education*, 31(4), 980–1021. <https://doi.org/10.1007/s40593-021-00249-z>
- Ferguson, P. (2011). Student perceptions of quality feedback in teacher education. *Assessment & Evaluation in Higher Education*, 36(1), 51–62. <https://doi.org/10.1080/02602930903197883>
- Fränti, P., & Mariescu-Istodor, R. (2023). Soft precision and recall. *Pattern Recognition Letters*, 167, 115–121. <https://doi.org/10.1016/j.patrec.2023.02.005>
- Fu, X., Befferman, T., Chiu, J., & Burghardt, M. D. (2015). Wisengineering: Achieving scalability and extensibility in massive online learning. In: J. Wang, W. Cellary, D. Wang, H. Wang, S. -C. Chen, T. Li, & Y. Zhang (Eds), *Web Information Systems Engineering WISE* (Vol. 9418, pp. 323–337). Springer, Amsterdam
- Garrison, D. R. (1995). Constructivism and the role of self-instructional course materials: A reply. *Distance Education*, 16(1), 136–140. <https://doi.org/10.1080/0158791950160110>
- Gasevic, D., Jovanovic, J., Pardo, A., & Dawson, S. (2017). Detecting learning strategies with analytics: Links with self-reported measures and academic performance. *Journal of Learning Analytics*, 4(2). <https://doi.org/10.18608/jla.2017.42.10>
- Gasevic, D., Dawson, S., & Siemens, G. (2015). Let's not forget: Learning analytics are about learning. *TechTrends*, 59(1), 64–71. <https://doi.org/10.1007/s11528-014-0822-x>
- Hadwin, A. F., Nesbit, J. C., Jamieson-Noel, D., Code, J., & Winne, P. H. (2007). Examining trace data to explore self-regulated learning. *Metacognition and Learning*, 2, 107–124. <https://doi.org/10.1007/s11409-007-9016-7>
- Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112.
- Hicks, S. A., Strümke, I., Thambawita, V., Hammou, M., Riegler, M. A., Halvorsen, P., & Parasa, S. (2022). On evaluation metrics for medical applications of artificial intelligence. *Scientific Reports*, 12(1), Article 5979. <https://doi.org/10.1038/s41598-022-09954-8>
- Jansen, T., Höft, L., Bahr, L., Fleckenstein, J., Möller, J., Köller, O., & Meyer, J. (2024). Comparing generative AI and expert feedback to students' writing: Insights from student teachers. *Psychologie in Erziehung und Unterricht*, 71(2), 80–92.
- Kaufman, L., & Rousseeuw, P. J. (2009). Finding groups in data: An introduction to cluster analysis. *John Wiley & Sons*. <https://doi.org/10.1002/9780470316801>

- Kizilcec, R. F., Pérez-Sanagustín, M., & Maldonado, J. J. (2017). Self-regulated learning strategies predict learner behavior and goal attainment in Massive Open Online Courses. *Computers & Education*, 104, 18–33. <https://doi.org/10.1016/j.compedu.2016.10.001>
- Kosmyna, N., Hauptmann, E., Yuan, Y. T., Situ, J., Liao, X.-H., Beresnitzky, A. V., Braunstein, I., & Maes, P. (2025). Your brain on chatgpt: Accumulation of cognitive debt when using an ai assistant for essay writing task. *arXiv*. <https://doi.org/10.48550/ARXIV.2506.08872>
- Kumar, H., Rothschild, D. M., Goldstein, D. G., & Hofman, J. M. (2025). Math education with large language models: Peril or promise? In: A. I. Cristea, E. Walker, Y. Lu, O. C. Santos, & S. Isotani (Eds.), *Artificial Intelligence in Education* (Vol. 15880, pp. 60–75). Springer, Switzerland
- Lajoie, S. P., & Azevedo, R. (2006). Teaching and learning in technology-rich environments. *Handbook of Educational Psychology*, 2, 803–821.
- Lan, M., & Zhou, X. (2025). A qualitative systematic review on AI empowered self-regulated learning in higher education. *Npj Science of Learning*, 10(1), Article 21. <https://doi.org/10.1038/s41539-025-00319-0>
- Lee, T. H., Shen, P. D., & Tsai, C. W. (2008). Applying web-enabled problem-based learning and self-regulated learning to add value to computing education in Taiwan's vocational schools. *Journal of Educational Technology & Society*, 11(3), 13–25.
- Lehmann, T., Hähnlein, I., & Ifenthaler, D. (2014). Cognitive, metacognitive and motivational perspectives on prefection in self-regulated online learning. *Computers in Human Behavior*, 32, 313–323. <https://doi.org/10.1016/j.chb.2013.07.051>
- Li, Q., Peng, H., Li, J., Xia, C., Yang, R., Sun, L., Yu, P. S., & He, L. (2022). A survey on text classification: From traditional to deep learning. *ACM Transactions on Intelligent Systems and Technology*, 13(2), 1–41. <https://doi.org/10.1145/3495162>
- Lim, L. A., Gasevic, D., Matcha, W., Ahmad Uzir, N., & Dawson, S. (2021). Impact of learning analytics feedback on self-regulated learning: Triangulating behavioural logs with students' recall. *LAK21: 11th International Learning Analytics and Knowledge Conference*. <https://doi.org/10.1145/3448139.3448174>
- Maloney, S., Axelsen, M., Galligan, L., Turner, J., Redmond, P., Brown, A., Basson, M., & Lawrence, J. (2022). Using lms log data to explore student engagement with coursework videos. *Online Learning*, 26(4), 399.
- Mangaroska, K., & Giannakos, M. (2019). Learning analytics for learning design: A systematic literature review of analytics-driven design to enhance learning. *IEEE Transactions on Learning Technologies*, 12(4), 516–534. <https://doi.org/10.1109/TLT.2018.2868673>
- Martin, E. L., & Martinez, D. A. (2023). The effect size in scientific publication. *Educación XXI*, 26(1), 9–17.
- Martin, H., Craigwell, R., & Ramjarrie, K. (2022). Grit, motivational belief, self-regulated learning (SRL), and academic achievement of civil engineering students. *European Journal of Engineering Education*, 47(4), 535–557. <https://doi.org/10.1080/03043797.2021.2021861>
- Meyer, J., Jansen, T., Schiller, R., Liebenow, L. W., Steinbach, M., Horbach, A., & Fleckenstein, J. (2024). Using LLMs to bring evidence-based feedback into the classroom: AI-generated feedback increases secondary students' text revision, motivation, and positive emotions. *Computers and Education: Artificial Intelligence*, 6, Article 100199. <https://doi.org/10.1016/j.caeai.2023.100199>
- Mojarad, S., Essa, A., Mojarad, S., & Baker, R. S. (2018). Data-driven learner profiling based on clustering student behaviors: Learning consistency, pace, and effort. In: *Intelligent Tutoring Systems: 14th International Conference, ITS 2018, Montreal, QC, Canada, June 11–15, 2018, Proceedings 14* (pp. 130–139). Springer
- Nazaretsky, T., Mejia-Domenzain, P., Swamy, V., Frej, J., & Käser, T. (2024). Ai or human? Evaluating student feedback perceptions in higher education. In R. Ferreira Mello, N. Rummel, I. Jivet, G. Pishtari, & J. A. Ruipérez Valiente (Eds.), *Technology Enhanced Learning for Inclusive and Equitable Quality Education* (Vol. 15159, pp. 284–298). Springer
- Ng, D. T. K., Tan, C. W., & Leung, J. K. L. (2024). Empowering student self-regulated learning and science education through ChatGPT: A pioneering pilot study. *British Journal of Educational Technology*, 55(4), 1328–1353. <https://doi.org/10.1111/bjet.13454>
- Pardo, A., Jovanovic, J., Dawson, S., Gašević, D., & Mirriahi, N. (2019). Using learning analytics to scale the provision of personalised feedback. *British Journal of Educational Technology*, 50(1), 128–138. <https://doi.org/10.1111/bjet.12592>
- Pintrich, P. R. (2000). The role of goal orientation in self-regulated learning. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 451–502). Elsevier.
- Reddy, B. H., & Karthikeyan, P. R. (2022, November). Classification of fire and smoke images using decision tree algorithm in comparison with logistic regression to measure accuracy, precision, recall, and

- F-score. In *2022 14th International Conference on Mathematics, Actuarial Science, Computer Science and Statistics (MACS)* (pp. 1–5). IEEE
- Sagr, M., López-Pernas, S., & Vogelsmeier, L. V. D. E. (2023). When, how and for whom changes in engagement happen: A transition analysis of instructional variables. *Computers & Education*, 207, Article 104934. <https://doi.org/10.1016/j.compedu.2023.104934>
- Sardi, J., Darmansyah, CandraO., & YulianaHabibullahEliza, D. F. D. T. P. Y. F. (2025). How generative ai influences students' self-regulated learning and critical thinking skills? A systematic review. *International Journal of Engineering Pedagogy (iJEP)*, 15(1), 94–108.
- Schmitz, B., & Wiese, B. S. (2006). New perspectives for the evaluation of training sessions in self-regulated learning: Time-series analyses of diary data. *Contemporary Educational Psychology*, 31(1), 64–96. <https://doi.org/10.1016/j.cedpsych.2005.02.002>
- Schunk, D. H., & Ertmer, P. A. (2000). Self-regulation and academic learning. İçinde *Handbook of Self-Regulation* (ss. 631–649). Elsevier
- Schunk, D. H. (2001). Self-regulation through goal setting. *The Handbook of Self-Regulation*, 1, 545–570.
- Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research*, 78(1), 153–189. <https://doi.org/10.3102/0034654307313795>
- Steinbach, M., Bhandari, S., Meyer, J., & Pardos, Z. A. (2025). When llms hallucinate: Examining the effects of erroneous feedback in math tutoring systems. *Proceedings of the Twelfth ACM Conference on Learning @ Scale*, 139–150. <https://doi.org/10.1145/3698205.3729555>
- Tang, H. (2021). Person-centered analysis of self-regulated learner profiles in MOOCs: A cultural perspective. *Educational Technology Research and Development*, 69(2), 1247–1269. <https://doi.org/10.1007/s11423-021-09939-w>
- Taylor, C., & Burke Da Silva, K. (2014). An analysis of the effectiveness of feedback to students on assessed work. *Higher Education Research & Development*, 33(4), 794–806. <https://doi.org/10.1080/07294360.2013.863840>
- Tsai, Y.-S., Mello, R. F., Jovanović, J., & Gašević, D. (2021). Student appreciation of data-driven feedback: A pilot study on OnTask. *LAK21: 11th International learning analytics and knowledge conference*, 511–517. <https://doi.org/10.1145/3448139.3448212>
- Van Alten, D. C. D., Phielix, C., Janssen, J., & Kester, L. (2021). Secondary students' online self-regulated learning during flipped learning: A latent profile analysis. *Computers in Human Behavior*, 118, Article 106676. <https://doi.org/10.1016/j.chb.2020.106676>
- Van Der Kleij, F. M., Feskens, R. C. W., & Eggen, T. J. H. M. (2015). Effects of feedback in a computer-based learning environment on students' learning outcomes: A meta-analysis. *Review of Educational Research*, 85(4), 475–511. <https://doi.org/10.3102/0034654314564881>
- Van Halem, N., Van Klaveren, C., Drachsler, H., Schmitz, M., & Cornelisz, I. (2020). Tracking patterns in self-regulated learning using students' self-reports and online trace data. *Frontline Learning Research*, 8(3), 140–163. <https://doi.org/10.14786/flr.v8i3.497>
- Verstege, S., Pijsira-Díaz, H. J., Noroozi, O., Biemans, H., & Diederens, J. (2019). Relations between students' perceived levels of self-regulation and their corresponding learning behavior and outcomes in a virtual experiment environment. *Computers in Human Behavior*, 100, 325–334. <https://doi.org/10.1016/j.chb.2019.02.020>
- Voelkel, S., Varga-Atkins, T., & Mello, L. V. (2020). Students tell us what good written feedback looks like. *FEBS Open Bio*, 10(5), 692–706. <https://doi.org/10.1002/2211-5463.12841>
- Weber, F., Wambsganss, T., & Söllner, M. (2024). Enhancing legal writing skills: The impact of formative feedback in a hybrid intelligence learning environment. *British Journal of Educational Technology*, 56(2), 650–677. <https://doi.org/10.1111/bjet.13529>
- Wiley, D. A. (2006). Learning objects in public and higher education. *Innovations in instructional technology* (pp. 1–9). Routledge.
- Ye, D. (2022). The history and development of learning analytics in learning, design, & technology field. *TechTrends*, 66(4), 607–615. <https://doi.org/10.1007/s11528-022-00720-1>
- Ye, D., & Pennisi, S. (2022). Using trace data to enhance students' self-regulation: A learning analytics perspective. *The Internet and Higher Education*, 54, Article 100855. <https://doi.org/10.1016/j.iheduc.2022.100855>
- Yot-Domínguez, C., & Marcelo, C. (2017). University students' self-regulated learning using digital technologies. *International Journal of Educational Technology in Higher Education*, 14(1), Article 38. <https://doi.org/10.1186/s41239-017-0076-8>
- Zheng, S., Rossom, M. B., Shih, P. C., & Carroll, J. M. (2015). Understanding student motivation, behaviors and perceptions in moocs. *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*, 1882–1895. <https://doi.org/10.1145/2675133.2675217>

- Zhou, J., Müller, H., Holzinger, A., & Chen, F. (2024). Ethical CHATGPT: Concerns, challenges, and commandments. *Electronics*, 13(17), Article 3417. <https://doi.org/10.3390/electronics13173417>
- Zimmerman, B. J. (2008). Investigating self-regulation and motivation: Historical background, methodological developments, and future prospects. *American Educational Research Journal*, 45(1), 166–183. <https://doi.org/10.3102/0002831207312909>

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